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Advances in Managing Chytridiomycosis for Australian Frogs: *Gradarius Firmus Victoria*

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Keywords

chytridiomycosis, amphibians, conservation, *Batrachochytrium dendrobatidis*, interventions

Abstract

Extensive knowledge gains from research worldwide over the 25 years since the discovery of chytridiomycosis can be used for improved management. Strategies that have saved populations in the short term and/or enabled recovery include captive breeding, translocation into disease refugia, translocation from resistant populations, disease-free exclosures, and preservation of disease refuges with connectivity to previous habitat, while antifungal treatments have reduced mortality rates in the wild. Increasing host resistance is the goal of many strategies under development, including vaccination and targeted genetic interventions. Pathogen-directed strategies may be more challenging but would have broad applicability. While the search for the silver bullet solution continues, we should value targeted local interventions that stop extinction and buy time for evolution of resistance or development of novel solutions. As for most invasive species and infectious diseases, we need to accept that ongoing management is necessary. For species continuing to decline, proactive deployment and assessment of promising interventions are more valid than a hands-off, do-no-harm approach that will likely allow further extinctions.

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INTRODUCTION

Chytridiomycosis is a pandemic that emerged after the fungal pathogens escaped from Asia, resulting in global declines in at least 501 amphibian species, including 90 extinctions (1). Chytridiomycosis is the worst disease to affect vertebrate biodiversity and requires a global effort to abate its impact (1, 2). Despite huge knowledge gains from 25 years of research since the discovery of chytridiomycosis (3–5), there is some pessimism around developing a sustainable solution (6). However, research has informed successful interventions that have saved species in the short term and/or at local scales, and promising approaches for long-term conservation progress are under development.

As with other invasive species and infectious diseases, a range of ongoing local measures can be used for control. Despite the current lack of a silver bullet that will broadly solve this issue forever, optimism and perseverance are needed, as well as an acceptance that chytridiomycosis may require ongoing targeted and local control efforts into the future. Due to the huge diversity in physiology and ecology of amphibian species, strategies optimized for specific species are worthy of support. Hence we have our title, *Gradarius Firmus Victoria*, meaning "taking small steps toward victory" (courtesy AFC Richmond).

Several studies on population recovery (7, 8) have suggested amphibians' capacity to evolve resistance. Further, connected natural environments, with some that buffer against the worst impacts of chytridiomycosis, can facilitate population persistence (8, 9) and likely support adaptation. Hence, we advocate for any methods that bolster population size and slow or halt declines that may buy time for adaptations or for development of novel solutions.

Australian frogs have been hit hard by the arrival of *Batrachochytrium dendrobatidis* (Bd), with rapid loss of tropical species and slower but similarly severe declines in temperate and alpine species. Here we collate recent progress on managing chytridiomycosis, presenting feasible approaches that are being implemented successfully on local scales, as well as current research on promising ideas for broader disease control. Although we refer to global findings, we do not aim to be comprehensive but highlight advances since previous reviews on managing chytridiomycosis (10–13), with a focus on our current research strengths and priorities for Australia.

HISTORY OF AUSTRALIAN RESPONSE

In Australia, Bd likely arrived around 1978 near Brisbane and spread north and south over the next two decades, causing population declines (2, 12). Because the mass mortalities went unobserved, the fungus was not discovered and implicated as the cause until 1998 (3), by which time 6 rainforest species had disappeared and many high-altitude forests no longer had frogs calling along streams. Although we had no further extinctions, chytridiomycosis (**Figure 1**; see the sidebar titled Pathogenesis of Chytridiomycosis) (14–19) remains a major threat to amphibians in Australia, with 6 species critically endangered and at least 30 more threatened (12, 20, 21). In the decades since the outbreaks began, 11 species that declined have begun to recover (20). Current severe impacts of habitat loss, climate change, and invasive species have been quantified (21, 22).

After the discovery of chytridiomycosis, Australia responded with management actions faster than other countries (5). However, this response was still much slower than would be typical for emerging infectious diseases of livestock or humans. Lack of a system to respond to biodiversity disease outbreaks meant following the lengthy procedure for dealing with environmental threats, which tend to be chronic and slower acting. In summary, chytridiomycosis was reported in 1998 and nominated and accepted as a key threatening process in 2000 and 2002, respectively. This stimulated production of a national threat abatement plan (TAP) in 2006 (reviewed in 2016 and in 2022) (23, 24). Although a TAP does not guarantee resourcing, it does give a statutory basis for

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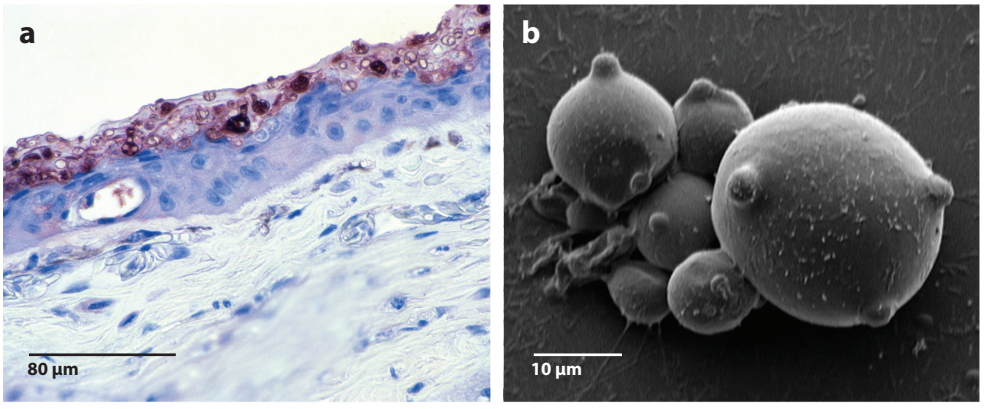

Figure 1

(*a*) An immunostained section of frog skin showing the red-stained sporangia of *Batrachochytrium dendrobatidis* in the superficial epidermis (adapted with permission from Reference 19). (*b*) Clump of sporangia in culture via scanning electron microscopy. Photo credit L. Berger.

decisions on Commonwealth funding. Additionally, zoos donated capacity early in the declines and have been able to maintain species that are now functionally extinct in the wild, such as the southern corroboree frog (*Pseudophryne corroboree*) (25).

In terms of research support for chytridiomycosis, substantial government resources were allocated initially to answer questions raised in the TAP. The urgency was recognized with funding provided via tender before the first TAP was formally approved, but funds decreased after 2008. For emerging infectious diseases in humans and livestock, the government leads strategic efforts for outbreak investigation, research, and management. In contrast, these responses to chytridiomycosis have often been driven by universities and zoos using short-term funding. Hence, research progress on improving management has been slowed by scientists needing to continually apply for resources and grants that are usually judged more on scientific than on conservation merit and that have low chances of being funded. Notable exceptions on management effort came from the

PATHOGENESIS OF CHYTRIDIOMYCOSIS

As the first species from the phylum Chytridiomycota found to cause disease in a vertebrate, *Batrachochytrium dendrobatidis* has presented many challenges for understanding and control. It grows intracellularly within the superficial epidermal layers, causing hyperkeratosis but minimal inflammatory reaction (14). Lack of inflammation could stem from the superficial intracellular location or due to secretion of immunosuppressant compounds (15). An early question on how this skin disease results in death was solved by studies showing that terminally infected frogs undergo a severe drop in blood electrolytes that leads to cardiac arrest, following from disruption of skin osmoregulatory functions and ion transport (16). Although the pathogen has a broad amphibian host range, species vary greatly in susceptibility. Resistance has been linked to skin peptide production, major histocompatibility complex genes, and the cutaneous microbiome, with evidence for acquired immunity in only some species (17, 18). Identifying resistance factors that enable some individuals to survive remains a key research topic.

In 2014, a second amphibian chytrid, *Batrachochytrium salamandrivorans*, was reported after arrival in Europe, where it decimated fire salamanders (*Salamandra salamandra*) (5). This fungus infects deep epidermal cells, resulting in a more ulcerative disease. It has not been detected in Australia.

extraordinary contributions of individual champions in state governments and nongovernmental organizations (e.g., 12, 22, 26).

The establishment of captive programs for nine species has prevented extinctions of some listed species. However, reintroductions continue to be sabotaged by high mortality due to chytridiomycosis, and the challenge remains to create self-sustaining wild populations. Successful interventions are described below, such as disease-free exclosures for *P. corroboree* and translocations of the armoured mist frog (*Litoria lorica*). Species given a lower priority for direct management of chytridiomycosis are those that have declined due to the disease but are persisting in smaller distributions and at lower abundance. Preserving habitat and reducing impacts of other threats are considered vital to supporting these species (12, 22).

UNDERSTANDING AND LIMITING FURTHER SPREAD

Genetic evidence identifies East Asia as the origin of Bd, which has spread worldwide due to the amphibian trade (27). A surprising amount of genetic diversity has been discovered in Bd, suggesting adaptation of certain genotypes to specific host species (28, 29). Sequenced Australian Bd strains all belonged to a single emerging lineage [global panzootic lineage, GPL (27)], supporting the epidemiological theory that it arrived via a rare human-assisted dispersal event into southeast Queensland (30, 31). Even in infected zones, the risk of introducing new strains, and their mycoviruses (32), must be minimized; consequences are hard to predict and could include creation of novel and more virulent Bd hybrids (33). Although Australia's unusually strict biosecurity for international arrivals and trade makes it illegal to import amphibians as pets and aims to control stowaways in produce, which reduces the risk of introducing novel strains, controlling spread of Bd GPL within Australia is also important.

Extensive mapping and predictive modeling indicate that Bd has spread throughout most of Australia's cooler and wetter habitats, where conditions are ideal for growth (30). However, a few suitable places with endemic frogs still appear to be negative, such as Cape York, Queensland (34); isolated desert and mountain sites; and parts of Southwest Tasmania (30). Biosecurity has been ramped up in Tasmania, and educational material has been generated to stop the spread of aquatic diseases including Bd (35). New biosecurity protocols and modifications to permitted recreational activities, such as moving campsites and stream access points downstream (in the case of Cape York), have been suggested to protect chytrid-naïve populations (34). Permits for research or other activities in protected areas are often made conditional on abiding by hygiene protocols to minimize chytridiomycosis transmission and spread (36).

Although the arid inland of Australia is generally inhospitable for Bd and appears to be disease free, there is potential for cool water bodies in the desert to maintain infections, as seen in Arizona and Nevada (37, 38). Hence, the risk to isolated endemic species needs addressing.

Research suggests that Bd may spread in infected soil and water and from the movement of infected hosts, especially via tolerant amphibian hosts acting as vectors. The common eastern froglet (*Crinia signifera*) has been identified as a competent vector with a high Bd prevalence, which increases infection risk in susceptible alpine species like the northern corroboree frog (*Pseudophryne pengilleyi*) and likely impacts reintroduction success in some areas (39–41). Non-amphibian hosts of Bd have been suggested [e.g., aquatic invertebrates (42)]. However, in Australia, Bd has yet to be detected in invertebrates, and a survey of crayfish was negative (43). Currently, knowledge gaps remain regarding how Bd persists and spreads and its potential for saprobic growth, as well as the importance of non-amphibian vectors. This knowledge could inform methods to reduce natural spread.

A recently reported issue is an introduced population of European smooth newts (*Lissotriton vulgaris*) in Victoria that was detected in 2011 and has an overall Bd prevalence of 6.3% (44). We

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are currently sequencing a fungal isolate from this population due to concerns that the newts may have brought in a new strain of Bd. The cane toad (*Rhinella marina*) was introduced to Australia in 1935 but is unlikely an important reservoir of Bd (45).

Surveys from Papua New Guinea (PNG), Australia's close neighbor, suggest that it has remained free of Bd despite having suitable habitats (46). Frogs of PNG are closely related to Australian species and are thus predicted to be highly susceptible (46). Therefore, disease preparedness and biosecurity are important to protect the diverse amphibians endemic to PNG.

An additional risk is the potential invasion of *Batrachochytrium salamandrivorans* into Australia and PNG. Although it primarily impacts salamanders, new studies show this more recently emerged chytrid is also pathogenic toward anurans (47), and we do not know how it will affect Australian frogs.

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HOST-DIRECTED INTERVENTIONS TO HALT BIODIVERSITY LOSS

Translocations

Releases and translocations can be successful despite the ongoing presence of Bd if environmental or host factors are exploited to promote coexistence. A successful translocation in Queensland involved *L. lorica*, which was predominantly a rainforest species that was believed to be extinct. However, one population was discovered persisting at a sunnier and drier site on the edge of the rainforest, where large boulders retained heat. This species was successfully translocated to a nearby site 4 km upstream with similar habitat features, and the reintroduced population has persisted and bred for more than 5 years (48, 49). Similarly, translocation of the spotted tree frog (*Litoria spenceri*) to a stream site with an open canopy aims to provide basking opportunities that reduce infection (22).

Another remarkable example is the reintroduction of mountain yellow-legged frogs (*Rana muscosa* and *Rana sierrae*) in California, where moving frogs from three recovering populations that had naturally evolved disease resistance enabled reestablishment in areas where they had disappeared (50). Analyses revealed evidence for genetic selection in the recovering populations that may explain this resistance. Hence, this is an example of evolutionary or genetic rescue, as discussed below.

This success in California supports the ideas Bell et al. (8) outlined for translocation of Queensland's Wet Tropics frogs from recovering populations to former sites. Although some natural recolonization to higher elevations occurred over two decades where there was habitat connectivity, with a gradual increase in disease risk, barriers were identified that could reduce spread of resistant frogs. For example, the waterfall frog (*Litoria nannotis*) could be translocated to unoccupied lowland sites at Big Tableland that are separated from extant populations by farmland.

A recent review of 15 translocation efforts of 7 Australian frog species that had declined from chytridiomycosis, using International Union for Conservation of Nature scoring, counted 3 (20%) translocations as highly successful, 7 (47%) as partially successful, and 5 (33%) as failures (51). Success was associated with efforts to reduce Bd impact at reintroduction sites, including increased salinity and selection of sites with warm microhabitats. This review provides a framework for considering approaches for translocation to reduce disease impacts, including host factors (e.g., age, resistance genotypes, vaccination) and environment (e.g., choice of natural habitats, manipulation of habitats, removal of reservoir species) (50). A plan for assessing assisted colonization sites that incorporates considerations for Bd and other threats has been developed for *P. pengilleyi* (41).

Captive Programs and Biobanking

Conservation breeding and biobanking can be used as stop gaps to prevent extinction and minimize genetic diversity loss. Conservation breeding has been effective at preventing extinction

from chytridiomycosis in many species in Australia and overseas [e.g., *P. corroboree*, Panamanian golden frogs (*Atelopus zeteki*)] (12, 25, 52). Captive colonies should be established with a goal of an exit strategy that addresses the threat of Bd in the wild and works toward understanding and promoting species resilience.

When possible, biobanking (cryopreservation of genetic material) should be performed on sperm or stem cells in addition to breeding to ensure preservation of founder population genetic diversity and composition (53, 54). So far, the only biobanking method that has generated viable offspring in amphibians is the use of sperm, whereas the larger size of eggs and embryos presents challenges to cryopreservation (53). In Australia, sperm cryopreservation methods are being developed for several priority threatened species, including *P. corroboree*, *P. pengilleyi*, *Litoria booroolongensis*, *Litoria castanea*, and *Litoria aurea* (54–56). Although technically challenging, somatic cell nuclear transfer has been used to restore extinct germplasm in threatened species such as black-footed ferrets and Przewalski's horse (57). This approach will likely be easier to perform in amphibians because they have large, externally fertilized eggs. It has been trialed for the de-extinction of the amazing southern gastric-brooding frog (*Rheobatrachus silus*) (58).

Genetic Intervention Strategies for Improving Host Resistance

Improving inherited host resistance to Bd is considered one of the most sustainable approaches to enabling frog survival and ending reliance on conservation interventions. The concept of genetic intervention is supported by field and laboratory studies indicating that resistance is heritable, and some recovering species have adapted to Bd after generations of persisting with the pathogen (7, 50, 59, 60). Current research on endangered frogs involves increasing resistance using selective breeding, genetic engineering, or genetic rescue. Many of these approaches can be adapted from methods proven in agriculture and aquaculture and may be needed for amphibians with populations so diminished that they have run out of time to adapt naturally in the wild.

Susceptibility to Bd varies between species, populations, and life stages (61), and studies have investigated the genetic basis of these differences. For example, genes associated with susceptibility have been identified in *P. corroboree* (62). Major histocompatibility complex (MHC) class II alleles were linked to resistance in Australian and Asian frogs, with evidence for selection in the wild (63).

There is growing evidence for natural selection of resistance. Recovering Australian species include Fleay's barred frog (*Mixophyes fleayi*), which inhabits the forest near the point of arrival of Bd in Australia in 1978 (64, 65). *M. fleayi* declined when Bd first emerged but appears to be evolving immunity, as indicated by its natural population recovery, with increased longevity associated with the ability of adults to limit and clear infection in the wild (65). In northern Queensland rainforests, some species that survived in the warmer lowlands after disappearing from their high-altitude cooler habitats have been gradually moving upward (8). Research on Central American rainforest frogs has linked population recovery to adaption of more effective skin peptides and not to a reduction in fungal virulence (7).

Efforts to increase or maintain genetic diversity, such as genetic rescue (e.g., through translocation), are likely to aid in species recovery through the impact of hybrid vigor (66). Genetic rescue has been applied successfully to increase fitness in several non-amphibian species, including mountain pygmy possums and European vipers (67–69). The benefits of genetic rescue are more likely in fragmented or inbred populations. Risks of genetic rescue include transmission of pathogens and other unintended impacts, such as outbreeding depression or genetic incompatibility (70–72).

In addition to increasing fitness by hybrid vigor, genetic rescue may introduce advantageous alleles (also referred to as evolutionary rescue) (66, 73, 74). Mendelson et al. (75) refer to this

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approach as a "recovery engine strategy" that attempts to increase Bd resistance by translocating Bd-tolerant frogs. This was successful in reestablishing populations of the mountain yellow-legged frog, described above (50). Genetic rescue may especially benefit populations that are suspected to differ in Bd susceptibility due to specific genes, such MHC allelic composition (62, 76). Genetic rescue may have also occurred accidentally (likely after anthropogenic transport) in a thriving population of northern leopard frogs (*Rana pipiens*) in the southwestern United States, which contains mitochondrial haplotypes of more Bd-resistant eastern populations of the species (77). This population with introduced haplotypes seems robust despite Bd-related declines in neighboring populations (77).

Targeted genetic intervention (TGI) is a more directed approach for increasing population resilience to intractable threats such as chytridiomycosis (78). TGI is a conservation tool kit that works by changing the incidence of alleles that impact fitness using methods such as artificial selection and synthetic biology (78). Although not yet trialed in a conservation context, TGI methods are well-established in agriculture and model organisms and have been used to increase disease resistance in several species, including flounder (79) and pigs (80).

TGI requires more resources than genetic rescue because it first requires characterizing the genetic basis of Bd resistance (i.e., heritability and genetic architecture). The genetic architecture can be characterized by performing Bd challenge studies to measure phenotypic variation in resistance that is correlated with genome-wide genetic data (e.g., a genome-wide association study) (81, 82). This knowledge informs selection of the best TGI approach, such as selective breeding or engineering of alleles that increase resistance (78). Synthetic biology approaches require intensive effort and testing to develop genetic engineering methods for non-model species (83) but could save species lacking any heritable Bd resistance. Artificial selection may be most effective in species with complex Bd resistance involving many genes. We are currently trialing TGI in Australia to increase Bd resistance in *P. corroboree* (**Figure 2**). The Vertebrate Genomes Project in the United States has contributed a reference genome of *P. corroboree* for this effort (84).

The best genetic intervention strategy for a population depends on characteristics including standing genetic diversity, risk of outbreeding depression or genetic incompatibility, heritability and genetic architecture of resistance, and intraspecific variation in resistance (70, 78, 85, 86). Any TGI approaches should be tested first in containment facilities to evaluate efficacy and any unintended impacts, followed by release of frogs into field enclosures and long-term post-release monitoring (78). Lastly, stakeholders impacted by the TGI process, including Indigenous people and local communities, should be involved in all steps of the decision-making process (78).

Figure 2

We are currently evaluating the heritability and genetic architecture of resistance to *Batrachochytrium dendrobatidis* (Bd) in southern corroboree frogs (*Pseudophryne corroboree*) in the laboratory by undertaking a disease challenge and genome-wide association study. The aim is to identify candidate Bd-resistance genes that will be used to test the efficacy of targeted genetic intervention. Photo credits (*left*) C. Doughty and (*middle*, *right*) T. Kosch.

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Vaccination

Initial studies suggested that effective acquired immunity was not stimulated by Bd infection or inoculations (17), but recent studies show "vaccinations" work in some species (18). Mild infections that were cured with heat or antifungals produced a response that promoted host resistance (18). Studies with leopard frogs from the United States (*Rana onca* and *R. pipiens*) showed that frogs had much reduced infections and greater survival after an experimental infection followed by antifungal treatment and then reexposure to the pathogen compared to those that were not exposed previously (18). Additionally, infection prevalence was reduced in Cuban tree frog (*Osteopilus septentrionalis*) tadpoles directly after exposure to metabolites from cultured Bd (87).

Although the resistance mechanisms are not yet understood, these methods of prior exposures or "vaccines" could be useful for boosting frog resistance before release. Because acquired immunological effects might not offer long-term protective immunity and are unlikely to be passed on to progeny, annual efforts to maintain resistance may be required. Initial trials of prerelease vaccination have had mixed success. Although laboratory vaccination trials of *Rana draytonii* in California reduced infection, these effects were not observed after release of frogs into the wild (88). More promisingly, protective effects of pre-exposure and clearance occurred in *L. aurea* in outdoor mesocosms (89). A high priority is to explore the use of acquired immunity in more species, as results are likely to be species specific.

Vaccinating wild frog populations will be difficult but could be facilitated by the development of transmissible vaccines. These could be developed using low-virulence strains of Bd or recombinant vaccines using viral vectors, similar to methods used against other wildlife diseases, e.g., rabbit myxomatosis (90). Laboratory trials have shown that low-virulence strains of Bd can spread between frogs and enable individuals to survive exposure to virulent strains for at least 34 weeks (18). In Europe, infecting frogs with low-virulence endemic Bd-GPL strains conferred protection against a more virulent GPL strain in the laboratory (91) and could also be protective in nature. However, impacts of transmissible vaccines on a host community in the wild may be complex and require thorough investigation before use (91).

REPRODUCTION AS A MECHANISM FOR POPULATION STABILITY

Reproductive responses to disease are often overlooked in favor of exploring immunity or environmental parameters that decrease disease. However, these mechanisms may be critical to how some species persist with endemic chytridiomycosis (61). In some persisting or recovering populations, there is little evidence of effective immunity reducing disease impacts; rather, increased reproduction via "terminal investment" might explain their survival (61, 92). The terminal investment hypothesis refers to the trade-off between investing in one large but final reproductive event before death versus investing in survival and future breeding. Increases in reproductive effort after infection have been demonstrated in various taxa, including house sparrows and limpets (93, 94).

In Australia, we have a prime example of a species persisting naturally after declines due to high annual recruitment, despite high annual mortality. Although they historically lived up to 7 years, alpine tree frogs (*Litoria verreauxii alpina*) now typically survive only until breeding at two years old, after which almost all adults die from chytridiomycosis (95–97). This puts populations at high risk of extirpation if drought dries their ephemeral breeding ponds, and persisting populations are now restricted to artificial permanent water bodies that support consistent annual recruitment (97).

Interestingly, disease seems to be directly impacting the reproductive effort of *L. verreauxii alpina*, because males appear to produce more gametes when infected (98). Increased reproductive effort may be evolving in populations with endemic disease, with evidence of baseline sperm production increasing over time (92). Also, frogs appear to be reaching maturity earlier (96).

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Figure 3

The alpine tree frog (*Litoria verreauxii alpina)* declined to approximately 20% of their former distribution in the Australian Alps. Populations currently appear stable and may be persisting with disease by increasing their reproductive output. Photo credit L. Brannelly.

Although annual recruitment is critical to the persistence of *L. verreauxii alpina* (92, 96) (**Figure 3**), it is not a stable long-term strategy. The Australian Alps are prone to extreme weather events such as fire and drought, making breeding inconsistent across years. However, experiments indicate *L. verreauxii alpina* may also be evolving an improved immune response after 20 years (99) despite high mortality in the wild each breeding season. This suggests there is potential for improved longevity of individuals despite the current reliance on annual breeding.

A high reproductive effort likely enables persistence with chytridiomycosis for other Australian species, such as *L. aurea* and *L. spenceri* (100, 101). If breeding is an important determinant of a species' survival with disease, then we should perhaps prioritize methods that promote recruitment, such as habitat manipulation or introduction to stable breeding sites.

ENVIRONMENTAL APPROACHES

Ecological studies have identified various environmental factors that influence disease outcomes and could be exploited, including increasing temperature and salinity, which have been reviewed comprehensively (13).

Refugia and Habitat Manipulation

Management strategies that preserve habitats that are suboptimal for Bd have become a priority for conservation. Ecological research has revealed that temperature is a key determinant of

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Figure 4

The common mistfrog (*Litoria rheocola*) declined in the Wet Tropics, Queensland, but is gradually regaining its former upland habitat. Photo credit L. Skerratt.

disease, and populations can persist when frogs can shelter in warm microclimates. Many species are persisting in the hotter and/or drier habitats within their former historic distributions (8, 13, 48, 102, 103). For example, populations living near thermal springs in Arizona have lower disease rates (104). In experiments, even short heat pulses can reduce infection burden (105). Water chemistry can also impact infection level; e.g., high salinity has been shown to improve frog survival in some species by lowering risk of transmission (9, 106–108). Habitats with a higher density of zooplankton that may predate on free-living stages of Bd have been linked to lower Bd prevalence (109). Hence, recognizing and preserving sites where frogs are persisting should be a priority (61), including impacted sites that may not be considered prime habitats, such as saline quarries that support growling grass frogs (*Litoria raniformis*) (9). Populations existing in refugia can act as a source population, enabling recolonization of connected habitats where species were extirpated [e.g., higher-elevation sites in the Wet Tropics (8, 103) (**Figure 4**)].

Habitat manipulation or creation to reduce disease impacts through increased temperatures or salinity has often been suggested and is supported by modeling (13, 107, 110, 111). Increasing salinity is promising for frogs that can withstand highly saline environments, such as *L. raniformis* and *L. aurea*. Releases of *L. aurea* into seminatural habitat with added salt had higher survival and lower infection for 18 months post release (112). Recent studies showed effective salt levels were unlikely to affect tadpole development and survival (106, 113).

Proposed manipulation techniques that increase temperature within a habitat include reducing canopy cover along streams (13, 107). A more direct approach, tested in a controlled outdoor mesocosm experiment, used brick piles passively warmed by the sun to provide heat shelters for frogs exposed to Bd (89) (**Figure 5**). *L. aurea* used these artificial habitats, resulting in greatly

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Figure 5

Green and golden bell frogs (*Litoria aurea*) sheltering in a warm brick pile, provided as a thermal refuge from chytridiomycosis. Photo credit A. Waddle.

reduced infection rates and loads through winter in Sydney, Australia, when chytridiomycosis epidemics are expected (89). Furthermore, frogs that self-cleared chytridiomycosis by selecting high temperatures were resistant to subsequent infections in outdoor mesocosms (89); hence, these artificial refugia may naturally promote acquired immunity within a population. The shelters are ready to be assessed in the wild in various climates.

The positive results of increased salinity and temperature in separate studies involving *L. aurea* suggest that combining these factors may be even more effective. However, it is important to monitor any interventions to assess their outcomes. For example, a recent laboratory study suggests higher UV exposure increases Bd loads in green tree frog (*Litoria caerulea*) tadpoles (114), possibly adding complexity to the interactions between management interventions and disease. Off-target impacts to be evaluated include effects on microbial, invertebrate, and vertebrate communities.

Bd-Free Exclosures

The New South Wales Government leads a remarkable effort to retain *P. corroboree* in the Southern Alps of Australia using large outdoor exclosures with metal walls that stop Bd transmission from frog species that act as reservoirs (e.g., *C. signifera*) (22). In these semi-wild habitats, frogs have remained uninfected, resulting in high survival in all life stages and successful reproduction. Because these frogs are subjected to many selective pressures from the wild, this approach preserves fitness characteristics that can be lost in captivity (22). These populations can generate frogs for release to the wild, providing opportunities for evolution of resistance when they are reintroduced to areas where Bd occurs. The efficacy of Bd-free exclosures should be investigated for other Australian species, particularly those with small home ranges.

ANTIFUNGALS AND DISINFECTANTS

Disinfectants used in simple environments could help control Bd in the wild, although off-target effects are a significant concern (115, 116). Many disinfectants have been trialed on Bd in the laboratory, but few have been proven safe to use on amphibians. A unique and successful trial in Mallorca, Spain, involved antifungal treatment and habitat sanitation to reduce infection in *Alytes muletensis* (115, 117). Tadpoles were collected and treated with the antifungal itraconazole while

the habitat was drained, dried, and disinfected with Virkon. The ponds refilled naturally, the frogs were returned, and these sites have remained Bd-free for at least seven years (115, 117). However, a similar application of draining and drying a pond (without chemical disinfection) in a related species (*Alytes obstetricans*) worked only in the short term, with infections reduced for one to two seasons (117). The habitat and ecology of the site in Mallorca appear ideal for Bd eradication, as *A. muletensis* was the only frog species present and sites are rocky and isolated. However, disease reintroduction events are much more likely to occur in most amphibian habitats, which are more complex (e.g., rainforests), making this mitigation approach unlikely to be broadly applicable.

The application of other disinfectants in natural systems has also been proposed. Peracetic acid was selected for study because it biodegrades quickly and has low toxicity for invertebrates, but safety is an issue unless all amphibians can be temporarily removed (116).

The antifungal agent epoxiconazole, which is applied on farms, was shown in the laboratory to bioaccumulate in newt skin and reach levels that inhibit Bd (118). This may help explain the minimal population-level effects of *Bd* in the agricultural western European region (118). Another potential disinfectant is F10, which is biodegradable and effective at extremely low concentrations (i.e., 7 15-min doses of 1:10,000 dilution) at disinfecting environments and curing infections (119).

The use of antifungal drugs like itraconazole to treat frogs in situ improved survival in the short term in multiple species (*A. muletensis*, *Leptodactylus fallax*, *Rana cascadae*, *R. sierrae*) (120–123). Itraconazole treatment can provide some protection against new infection for several weeks to several months. Knapp et al. (123) conducted five field experiments over a decade in mountain yellow-legged frogs (*R. sierrae*) in California and demonstrated that itraconazole treatment of tadpoles and subadults reduced Bd loads and improved survival, although infections returned to former levels within one year. However, survival of adults treated with itraconazole was higher over a three-year study, suggesting protective adaptive immunity occurred. Unfortunately, because early life stages remained susceptible, recruitment was limited and population declines continued.

NOVEL PATHOGEN-FOCUSED METHODS

Targeting the fungus could provide broadly applicable control methods but faces the challenges of pathogen evolution and balancing the reduction in virulence with fungal host fitness. For example, if fungal reproduction and growth are reduced, fungal strains may be outcompeted by more transmissible wild types.

Natural differences in virulence between Bd strains have been well-established (5, 124). Additionally, the virulence of isolates can be manipulated in the lab, with strains of higher passage history generally attenuating (125). Research on Australian isolates may identify low-virulence strains that could be exploited to outcompete more virulent strains, develop vaccines, or identify virulence factors.

Horticultural science has developed novel methods to target fungal virulence because crops are frequently plagued by fungal disease. RNA interference (RNAi) and the application of mycoviruses are two methods employed against plant fungal pathogens. Because these approaches are more specific than fungicides, there is less potential for off-target effects. Here we review preliminary research aiming to exploit these ideas to combat Bd.

RNA Interference

We have preliminary evidence that RNAi can reduce target gene expression in Bd (126) and are currently conducting trials to establish the feasibility of this approach for decreasing Bd virulence. RNAi is a natural cellular pathway that serves as a defense mechanism and gene-expression regulator in eukaryotes (127), and which can be manipulated to reduce fungal virulence. In fungi,

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RNAi occurs when foreign double-stranded RNA in the cytoplasm is cleaved into small fragments termed small interfering RNA (siRNA) (128). The siRNA is loaded onto an *Argonaute*-containing complex (129), which then guides the destruction of any corresponding messenger RNA, resulting in reduced gene expression, or gene knockdown. The components of RNAi are highly conserved across fungi, but several taxa lack some, or all, of the machinery (130). The Bd genome contains the necessary *Argonaute* and *Dicer* genes, although the presence of the specific RNA polymerase for amplification is unclear (131, 132). Additionally, a recent study found that Bd in tolerant hosts had upregulated RNAi genes compared to Bd in susceptible hosts (132), suggesting RNAi might be occurring in this system as a host defense against chytridiomycosis.

Synthetic RNA can be designed to exploit the fungal RNAi system and knock down gene expression, but a challenge is choosing genes to target to reduce virulence. Transcriptomic and genomic studies in Bd suggest that proteases such as serine-type and metalloproteases are good candidates because they play a role in germ tube invasion (131, 133). However, knockdown of proteases could be compensated by several alternative Bd protease genes. Ideal RNAi targets are nonredundant genes, such as the enzymes for antioxidant, polyamine, or metabolite synthesis.

Application of RNAi in plants typically involves spraying the leaves, as fungi naturally take up exogenous RNA from the environment (134) or from their plant hosts (135). Similarly, frog skin could be sprayed during seasons of high susceptibility. Frogs could be sprayed when they are easy to find, such as during calling, amplexus, or egg guarding. Once RNAi techniques are developed, the RNA code can be altered easily, allowing changes if the pathogen evolves resistance. However, like antifungal drugs, the effect of RNAi is not long lasting, requiring regular application.

Although topical siRNAs have not been used in animals, a similar but more enduring approach was effective against viruses in a range of vertebrates, such as foot-and-mouth disease in mice (136). In these cases, host-induced gene silencing was used via transgenesis to create hosts that express the interfering RNA.

Mycoviruses

Another horticultural technique being investigated as a self-sustaining solution is virus biocontrol. Mycoviruses are common obligate fungal parasites and have been used to reduce virulence in pathogenic fungi, such as those used to control plant fungal disease in crops, timber, and fruit trees (137). The most famous example of mycovirus biocontrol is the treatment of chestnut blight that has decimated European and US chestnut trees. An RNA mycovirus (CHV-1) was shown to reduce fungal virulence and was applied by simply inoculating diseased trees with the infected isolate (138). This treatment also reduced mortality in nearby trees, suggesting natural dispersion (138).

The first step toward using this approach to mitigate Bd is finding a suitable mycovirus that reduces Bd virulence. Surveys did not detect RNA mycoviruses in Bd isolates collected from throughout Australia ($n = 31$), the Americas ($n = 14$), and South Korea ($n = 2$), using conventional double-stranded RNA cellulose chromatography and transcriptome mining (139, 140).

The DNA mycovirus BdDV-1 was discovered recently in the genome of multiple Bd isolates (32). This mycovirus was common in endemic lineages (Bd-CAPE, Bd-ASIA1, Bd-Brazil) but was detected in only 1 of 211 sequences of Bd-GPL that had matches to viral genes. Hence, it also appears unlikely to occur in Australia, where only the GPL has been detected (27). Phenotypic differences were detected between naturally infected and uninfected Bd-Brazil isolates. Although viral presence was associated with decreased fungal growth in vitro, mycovirus infection increased virulence in amphibian infection trials, as indicated by higher host mortality rates. Hence, this virus appears unsuitable for biocontrol efforts but might be useful as a vector for genetic transformations of chytrids (32). Another consideration about this discovery is the possible risk of hybridization between lineages transferring BdDV-1 into a GPL strain (32).

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Because Bd likely had only a single introduction into Australia, this could explain the lack of RNA mycoviruses (140). Continued efforts to screen Bd for suitable RNA mycoviruses should focus on screening Bd from areas of high diversity, such as Asia (27), as well as specific isolates found to display lower virulence. Freeze/thawing, as well as repeated passing, can cure fungi of their mycoviruses (141), so freshly isolated fungal cultures should be used.

CONCLUSIONS

Chytridiomycosis is one of the most widely studied wildlife diseases, and there is cause for optimism for amphibian conservation due to the diverse strategies being developed using this knowledge base (11–13, 142). We can also adapt proven methods developed for agriculture, such as artificial selection. Novel proactive solutions are needed, because a hands-off approach will let more species disappear even from protected environments. We have summarized diverse management interventions: Some have proven success, some require assessment in wild frogs, and others are still in the early development stages.

Although reestablishing endangered species in the wild remains challenging, various local interventions have had short- or long-term success and can be implemented and assessed for further species. Many of these promising management strategies involve ongoing actions (e.g., habitat manipulation), but enabling species to persist provides them a chance to develop their own long-lasting survival mechanisms, such as evolution of resistance. Furthermore, for many invasive species, such as weeds, we accept that ongoing management is needed and is resourced.

Effective local field interventions included translocations, sourcing frogs from resistant populations or moving frogs to environmental refugia; creation of Bd-free exclosures; and elimination from a site using disinfectants (22, 48, 50, 115). Ecological studies have revealed frogs are persisting in locations that may be considered outside their traditional favored habitat, such as drier habitats or saline sites that are suboptimal for chytrid. Management that preserves or creates these habitats, including the general approach of conserving sites wherever frogs are thriving, must be prioritized, especially when outside protected environments (61).

Supporting persistence in the face of Bd includes mitigating any manageable threats where possible (22, 100, 143). This can include providing permanent water that enables breeding through dry years. Habitat that supports consistent annual recruitment has become crucial for species with high turnover due to Bd (96).

Improving host resistance is a priority for many critically endangered species. Immunization has been effective in outdoor mesocosms (89), and trials should be extended to free-ranging frogs and more species. Current research on TGIs aims for long-term change in disease resistance, including selective breeding for heritable resistance phenotypes or use of synthetic biology to confer genetic resistance. Pathogen-focused methods to reduce virulence are under early stages of development but have potential for broad applicability across host species.

The diverse biology of amphibians means there is unlikely to be one solution for managing Bd, and species-specific approaches are valuable. For many aspects of chytridiomycosis (e.g., vaccination success), conclusions cannot be extrapolated between amphibian species with certainty, and studies targeted to each species are required. Using successful strategies from other systems, however, is a sensible starting point.

For many frog species, chytridiomycosis has been their sole threat, uncomplicated by multiple factors and vested interests. However, as climate change progresses, many habitats are becoming less suitable for frogs (143), and predictions show severe impacts on Australian frogs without overall reductions in impacts of Bd (144). But mitigating chytridiomycosis will certainly increase population resilience to overcome other threats, such as coping with decreased recruitment during drought years (22).

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Progress on researching and managing chytridiomycosis has occurred despite many overarching challenges, including lack of resourcing for amphibians compared with mammals, as well as the general difficulties in funding applied conservation research. In contrast to many current conservation issues in Australia (e.g., forest clearance, feral horses), for which the main limitation is political, dedicated research is needed to find solutions to mitigate chytridiomycosis. This includes assessing interventions in the field within a controlled and adaptive framework. In Australia, a national chytridiomycosis research and management strategy with allocation of government funds to ensure a set of key priorities are addressed (12) would be more efficient than the current competitive and more random funding model. In conclusion, progress on various facets of managing chytridiomycosis has contributed toward the end goal of enabling self-sustaining populations, and we feel hopeful of achieving this ultimate conservation aim as part of the dedicated global effort.

DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

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